

Models for ants

In the absence of interactions, antineutrinos are not Braggian particles

Model for non-interacting ants: active Brownian particle

- Particle i follows
$$\begin{cases} dX_t^i &= \text{Pe} v(\theta_t^i) dt + \sqrt{2\sigma_x} dW_t^i \\ d\theta_t^i &= \sqrt{2} dB_t^i \end{cases}$$
- $\text{Pe} > 0, \sigma_x \geq 0$
- $v(\theta) = (\cos(\theta), \sin(\theta))^T$
- W_t^i 2D Brownian motion, B_t^i 1D Brownian motion



Model for ants

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- Model for non-interacting ants: active Brownian particle

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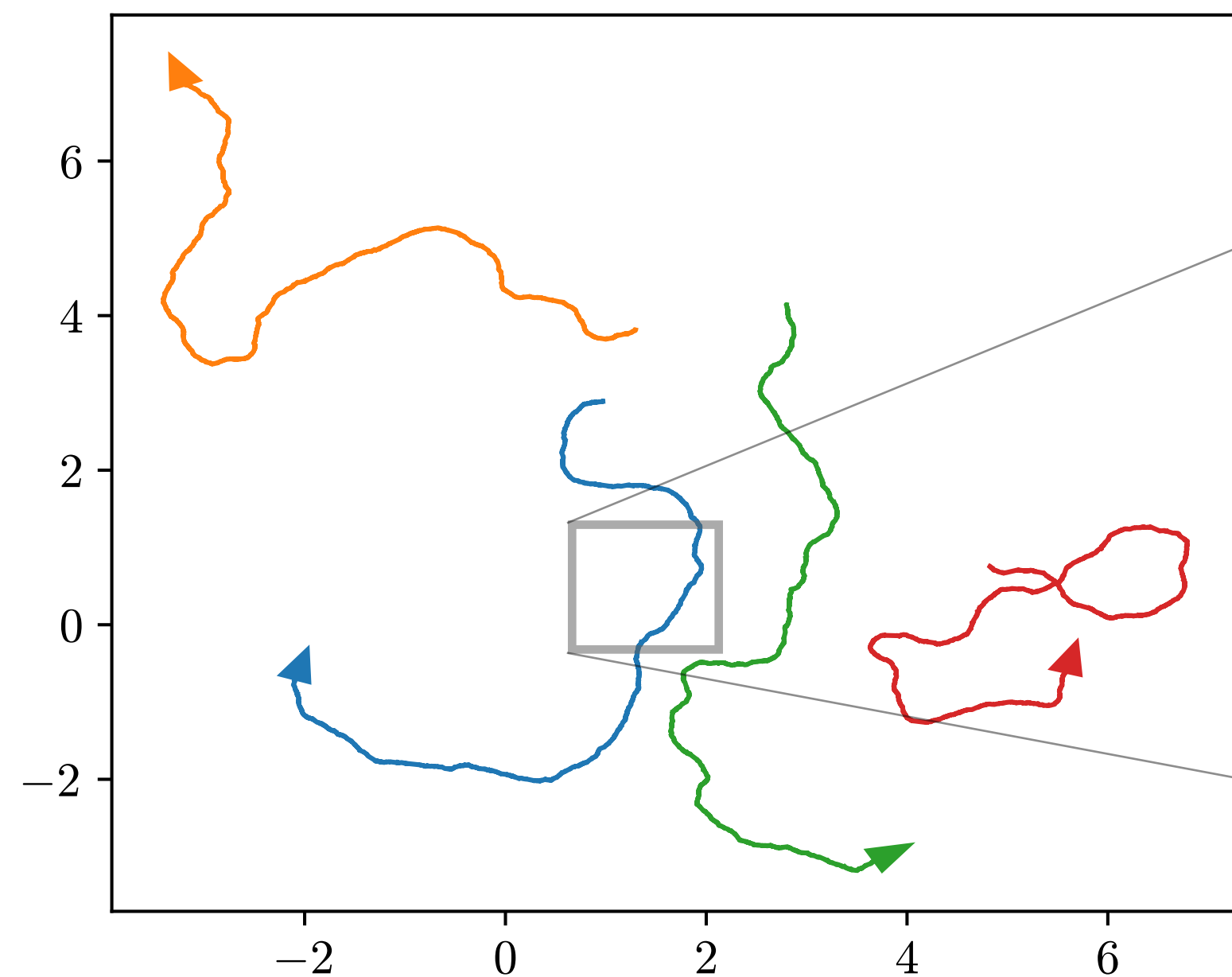
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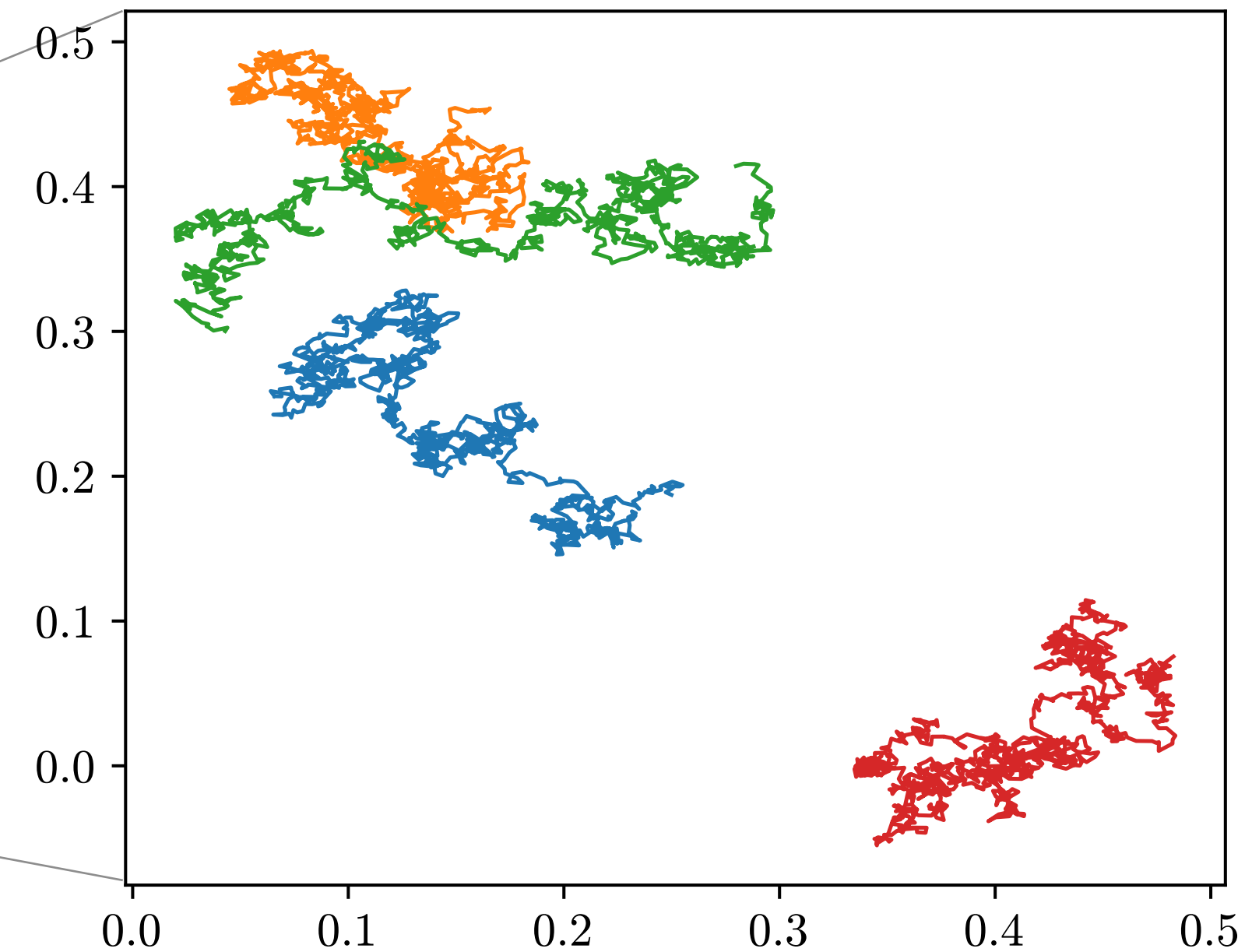
Model for ants

- 2D Active Brownian particles

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$\text{Pe} = 2, \sigma_x = 10^{-3}$



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